

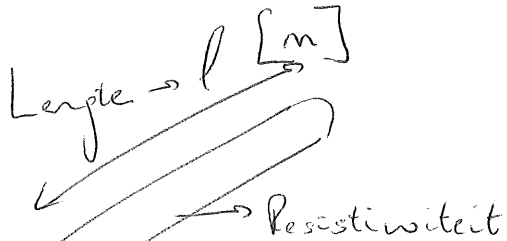
Hoofdstuk 2.

Lesing 2.1

①

* Ohm se Wet.

→ Weerstand: eienskap van materiaal wat vloei van lading keer.



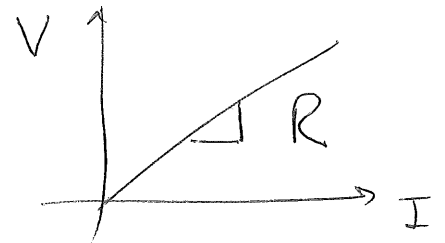
$$R = \rho \frac{l}{A}$$

* Resistiwiteit: Ω/m

→ Ohm se Wet:

spanning oor 'n weerstand is
direk eweredig aan stroom deur
die weerstand.

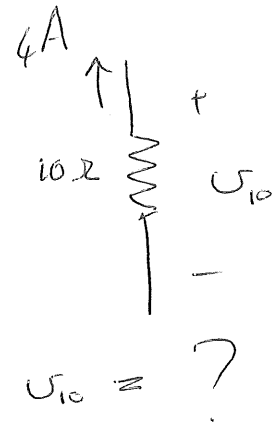
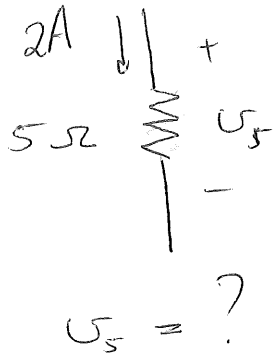
$$V = I \times R$$



Weerstand → Ω → Ohm.
(Omega)

* Tekentransversie vir Ohm

(2)



→ Stroom vloei van hoe energie na lae energie.

∴ Vloei van hoe spanning na lae spanning.

$$U_5 = +iR$$

$$= 2 \times 5$$

$$= 10V$$

$$U_{10} = -iR$$

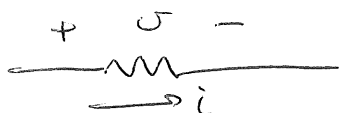
$$= -4 \times 10$$

$$= -40V.$$

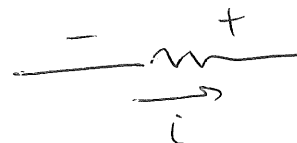
* Ohm se wet

$$U = \pm iR \quad \text{of} \quad i = \pm U/R$$

→ Spanning terminaal waar stroom by weerstand in vloei, bepaal teken.

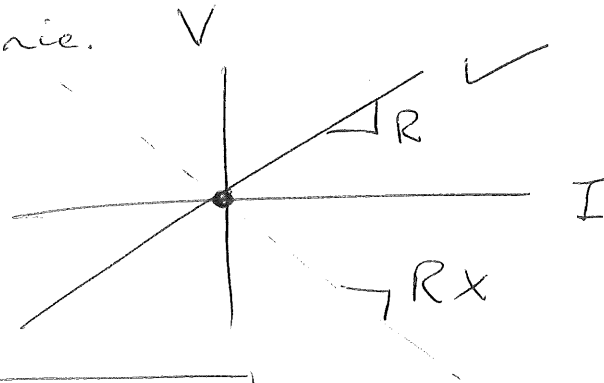


$$U = +iR$$



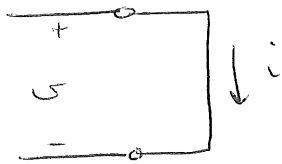
$$U = -iR.$$

Weerstandswaarde kan nooit negatief
wees nie.



$$|R| \geq 0$$

⇒ Wat gebeur by ekstreme waardes?
 0Ω $\infty \Omega$

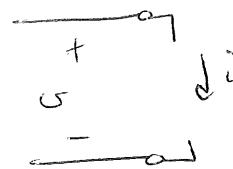


$$\begin{aligned} V &= i \times R \\ &= i \times 0 \\ &= 0V \end{aligned}$$

$$i = ? \neq 0!$$

→ Geen potensiaalverskil.

→ Pad vir stroom om
te vloei



$$\begin{aligned} i &= V/R \\ &= V/\infty \\ &= 0A \end{aligned}$$

$$V = ? \neq 0!$$

→ Geen pad vir
stroom nie.

→ Daar is wel 'n
potensiaalverskil.

* Geleiding (Conductance)

(4)

* Inverse van weerstand.

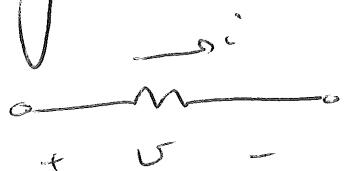
* Eenheden: S (Siemens)

Ω (mho)

$$G = \frac{1}{R} = \frac{i}{\omega}$$

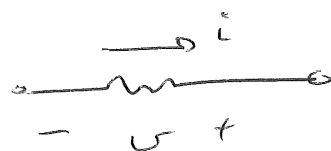


* Drywing voor Weerstand.



$$\begin{aligned} p &= +ui \\ &= iR \times i \\ &= i^2 R \end{aligned}$$

$$\left(\Delta \quad \begin{aligned} p &= +ui \\ &= +\frac{u^2}{R} \end{aligned} \right)$$

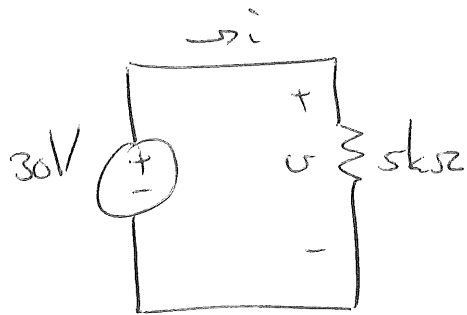


$$\begin{aligned} p &= -ui \\ &= -(-iR)i \\ &= i^2 R \end{aligned}$$

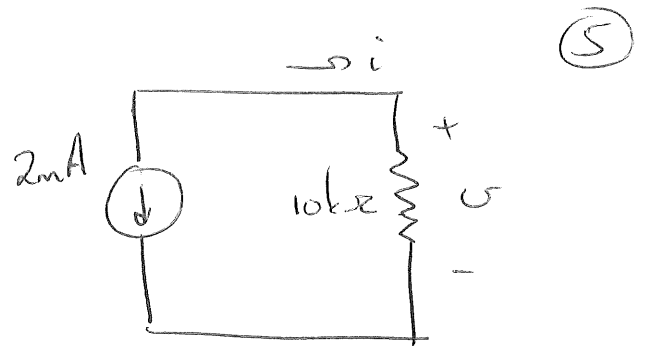
$$\left(\Delta \quad \begin{aligned} p &= -ui \\ &= -u\left(\frac{-u}{R}\right) \\ &= \frac{u^2}{R} \end{aligned} \right)$$

* Weerstand (resistor) sal altijd drywing absorbeer.

VL



$v = ?$
 $i = ?$
 $G = ?$
 $p = ?$



$v = ?$
 $i = ?$
 $G = ?$
 $p = ?$

⑤

Nodusse, Takke en Lusse (Nodes, Branches and Loops)

① Tak: enkel element in kringbaan.
↓
Bronne ↓ Resistors
 ↓
 Kapasitors

② Node: koppeling tussen twee of meer elemente.

③ Lus: enige geslote pad vir stroom om te vloei.

* Onafhanklik: lus wat nie gevorm kan word deur die kombinasie van ander lusse nie.

V6:

⑥

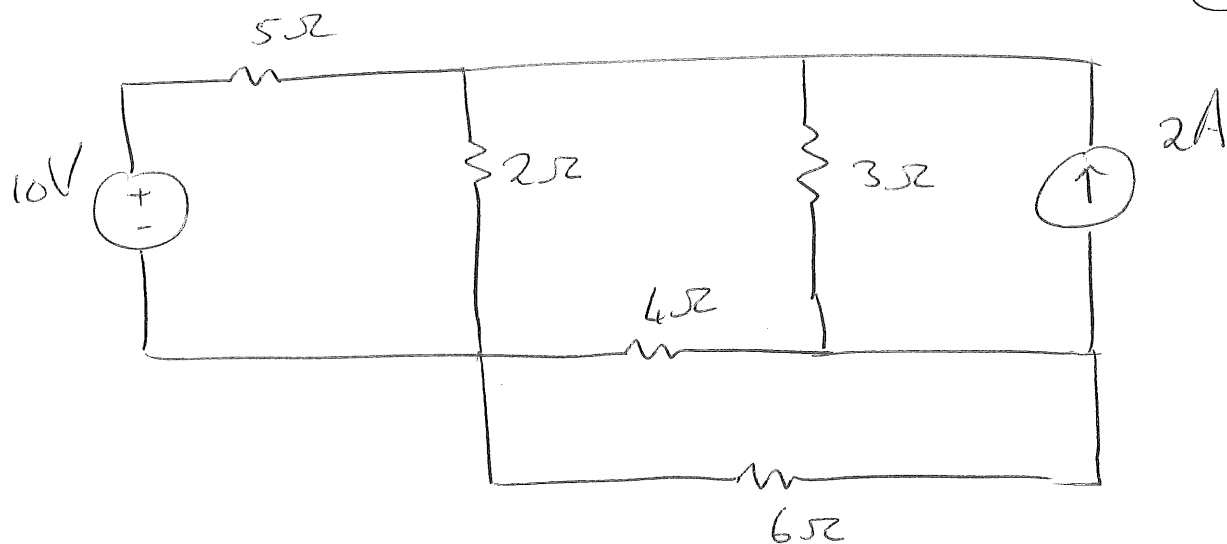


Table =

Nodusse =

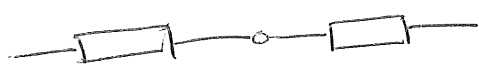
Onafh. Lusse =

$$\# T = \# L + \# N - 1$$



* Series + Parallel.

→ Series: ① twee elemente deel 1 node
eksclusief!



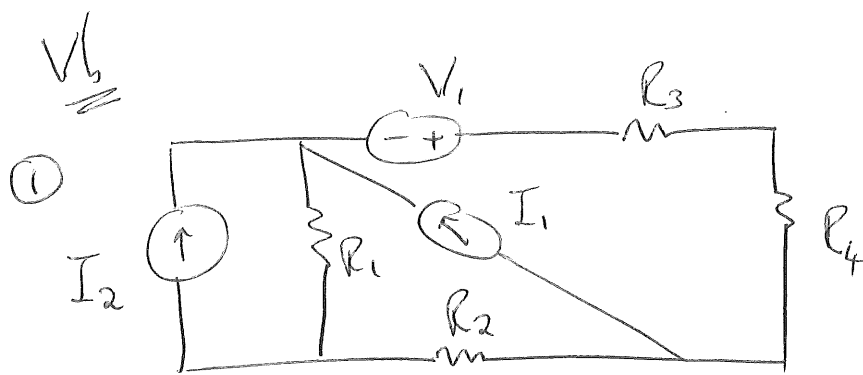
② ∴ dra dieseltdde stroom.

→ Parallel: ① twee of meer elemente deel
dieseltdde twee nodusse aan
weerstande van die element.



② ∴ deel dieseltdde spanning.

(7)



$$T = ?$$

$$N = ?$$

$$\text{onath. } L = ?$$

② What is in serie en wat is in parallel?

